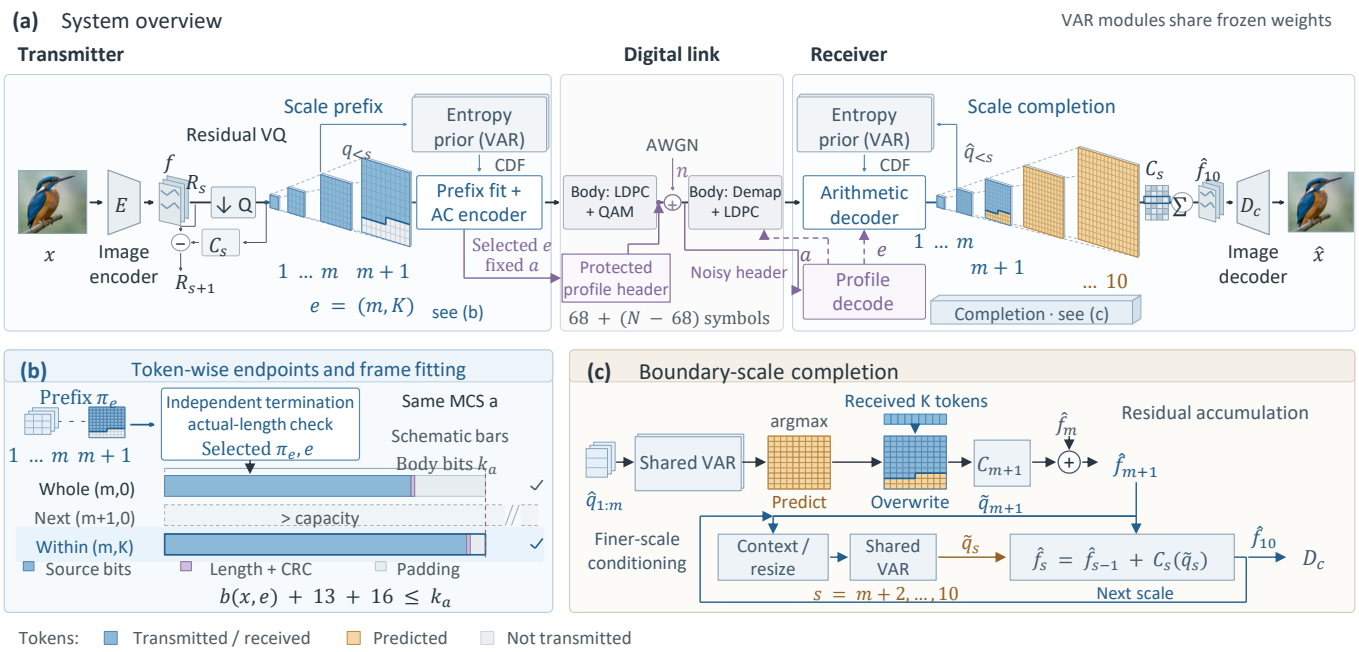

(a) System overview
VAR modules share frozen weights
Transmitter
Digital link
Receiver
Scale prefix
Residual VQ
Entropy prior (VAR)
CDF
Prefix fit + AC encoder
Selected e fixed a
AWGN
Body: LDPC + QAM
Body: Demap + LDPC
Protected profile header
Noisy header
Entropy prior (VAR)
CDF
Arithmetic decoder
Profile decode
Scale completion
Completion · see (c)
Image encoder
Image decoder
1 … m m + 1
e = (m, K)
see (b)
68 + (N − 68) symbols
… 10
(b) Token-wise endpoints and frame fitting
Prefix π_e
Independent termination actual-length check
Selected π_e, e
Same MCS a
Schematic bars
Body bits k_a
Whole (m,0)
Next (m+1,0)
> capacity
Within (m,K)
Source bits
Length + CRC
Padding
$b(x,e) + 13 + 16 \le k_a$
(c) Boundary-scale completion
Received K tokens
argmax
Predict
Overwrite
Shared VAR
Residual accumulation
Finer-scale conditioning
Context / resize
Shared VAR
$s = m+2, \ldots, 10$
Next scale
Tokens:
Transmitted / received
Predicted
Not transmitted